

MMD Generative Adversarial Networks

Meeting 3

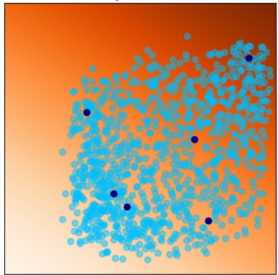
June 1, 2026

- 1 Analysis of stability of MMDGAN Training in 2d examples , Hyperparameters effects
- 2 Kernel choice effect
- 3 Convexity of the target distribution
- 4 MMD GANS training on CIFAR

Different values of σ in the RBF Kernel

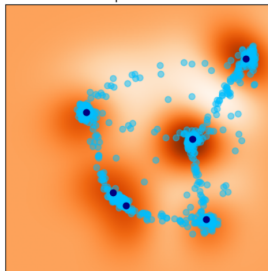
$$\sigma = 2$$

Epoch 295



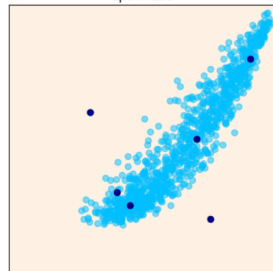
$$\sigma = 0.1$$

Epoch 295



$$\sigma = 0.005$$

Epoch 295



The MMD/Kernel Gradient

In an MMD GAN, the generator updates its parameters based on the gradient of the MMD distance. For a Gaussian Radial Basis Function (RBF) kernel:

$$k(x, y) = \exp\left(-\frac{\|x - y\|^2}{2\sigma^2}\right)$$

The gradient of this kernel with respect to a generated point x scales directly with the bandwidth parameter σ :

$$\nabla_x k(x, y) = -\frac{1}{\sigma^2}(x - y) \exp\left(-\frac{\|x - y\|^2}{2\sigma^2}\right)$$

Kernels of the type $k(x, y) = f(\|x - y\|^2)$

$$\nabla_x k(x, y) = 2(x - y)f'(\|x - y\|^2)$$

Kernels of the form $k(x, y) = f(\|(x - y)\|^2)$

Positive definite kernels of the form $k(x, y) = f(\|(x - y)\|^2)$

A kernel of the form $k(x, y) = f(\|(x - y)\|^2)$ is positive definite if and only if $f(\mathbf{R}^+) \subset \mathbf{R}_*^+$

f strictly decreasing on $\mathbf{R}^+$

Let $f : \mathbf{R}^+ \rightarrow \mathbf{R}_*^+ \in C^2(\mathbf{R}^+)$, $f(0) = 1$ strictly decreasing on $\mathbf{R}^+$

We suppose here that $f''(\mathbf{R}^+) \subset \mathbf{R}^+$

We define the d -bandwidth of f as $\sigma = \inf\{x | f'(x) < -d\}$

(Tradeoff bandwidth/Discriminative power)

Under the above assumptions:

$$\sigma d \leq 1$$

The Small Bandwidth Regime ($\sigma \rightarrow 0$)

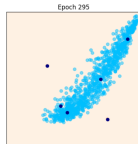
Vanishing gradient due to the small bandwidth

$$\lim_{\sigma \rightarrow 0} \nabla_x k(x, y) = 0 \quad \text{for } x \neq y$$

The generator receives **zero learning signal** guiding it toward the target distribution, completely freezing the learning efficiency.

- **Super-Exponential Decay:** The term $\exp\left(-\frac{\|x-y\|^2}{2\sigma^2}\right)$ dominates the gradient calculation and drops to zero at an extreme rate.
- **Flat Optimization Landscape:** The loss surface becomes a collection of isolated plateaus separated by wide, flat valleys.

$$\sigma = 0.005$$



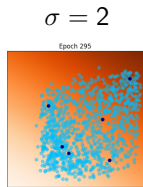
The Large Bandwidth Regime ($\sigma \rightarrow \infty$)

Uniformity caused by the large bandwidth

$$\nabla_x k(x, y) \approx -\frac{1}{\sigma^2}(x - y) \rightarrow 0 \quad \text{as } \sigma \rightarrow \infty$$

The gradient signal becomes uniform and incredibly weak. The network only learns the global mean, leaving the generator unable to capture complex, high-frequency details.

- **Loss of Discriminative Power:** The exponential term $\exp\left(-\frac{\|x-y\|^2}{2\sigma^2}\right) \rightarrow 1$, making the kernel blind to localized geometric differences.
- **Vanishing Scaling Factor:** The coefficient $\frac{1}{\sigma^2}$ shrinks the magnitude of the global gradient vectors down to near zero.



Transporting White noise into a discrete probability measure using a neural network

Theorem (Brenier's Theorem)

Let μ and ν be two probability measures on $\mathbb{R}^d$ with finite second moments, i.e.,

$$\int_{\mathbb{R}^d} \|x\|^2 d\mu(x) < \infty \quad \text{and} \quad \int_{\mathbb{R}^d} \|y\|^2 d\nu(y) < \infty.$$

Assume further that μ is absolutely continuous with respect to the Lebesgue measure.

Then, there exists a μ -essentially unique optimal transport map $T : \mathbb{R}^d \rightarrow \mathbb{R}^d$ that pushes μ forward to ν ($T_{\#}\mu = \nu$) minimizing the quadratic cost:

$$\min_{T_{\#}\mu = \nu} \int_{\mathbb{R}^d} \|x - T(x)\|^2 d\mu(x).$$

Furthermore, this optimal map T is uniquely characterized as the gradient of a convex function

$\Phi : \mathbb{R}^d \rightarrow \mathbb{R} \cup \{+\infty\}$:

$$T(x) = \nabla \Phi(x) \quad \mu\text{-almost everywhere.}$$

Learning an MMD-GAN on CIFAR (170M images)

An example of Results of the traing so far:

The blurry results might indicate a low value of the dimension of the latent space

Generated samples — epoch 40

